

During protein folding, which of the following interactions stabilize secondary, tertiary, or quaternary structures?

I. Hydrogen bonds

II. Ionic bonds

III. van der Waals forces

IV. Peptide bonds

☐ A. I and II only [6%]

☐ B. III and IV only [1%]

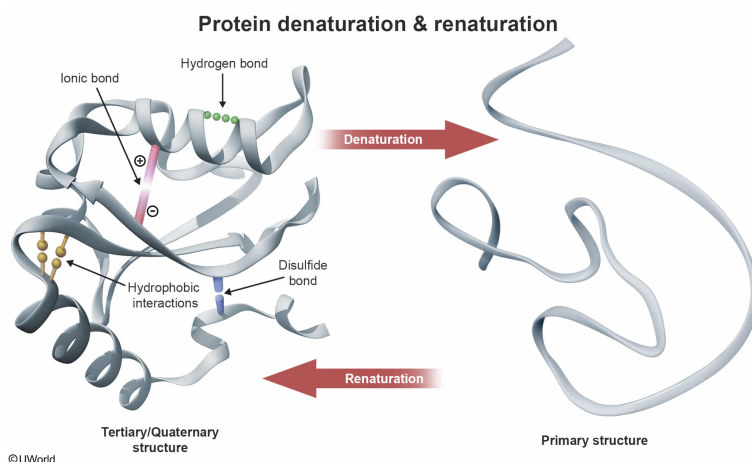
✓ ☒ C. I, II, and III only [73%]

☐ D. I, III, and IV only [19%]

Correct

73% answered correctly

## Explanation:



Proteins are polypeptides with varying levels of **structure**. Various chemical interactions underly the **secondary, tertiary, and quaternary structures** of a native protein, including hydrogen bonding, ionic bonding (tertiary and quaternary structures only), and van der Waals forces (**Numbers I, II, and III**).

**Secondary structure** is driven and reinforced by **hydrogen bonding** between the backbone carbonyl oxygen and the backbone amide hydrogen of different amino acids. Hydrogen bonds bend the polypeptide into  **$\alpha$ -helices and  $\beta$ -pleated sheets**, the main components of secondary structure.

**Tertiary structure** is formed by the folding of  $\alpha$ -helices and  $\beta$ -pleated sheets into a **three-dimensional** structure. Although this folding is driven by hydrophilic and hydrophobic interactions, the resulting structure is reinforced by **van der Waals forces, hydrogen**

**bonding** between side chains, and **ionic bonds** (electrostatic interactions or "salt bridges") formed between acidic and basic side chains. Tertiary structure may also be reinforced by **disulfide bonds (–S–S–)**, which are covalent bonds between thiol groups (–SH) of two cysteine side chains.

**Quaternary structure** refers to the assembly of two or more polypeptide chains (subunits) and is reinforced by the same chemical interactions as tertiary structure.

**(Number IV)** Peptide bonds are the **covalent linkages** between amino acids. They are created following the **condensation** of the  $\alpha$ -amino group of one amino acid with the  $\alpha$ -carboxyl of the second, forming an **amide** linkage. Peptide bonds stabilize the **primary structure** (ie, the linear sequence of amino acids) of a protein.

### **Educational objective:**

Primary protein structure is maintained by peptide bonds.

Secondary structure ( $\alpha$ -helices and  $\beta$ -pleated sheets) is maintained by hydrogen bonds. Tertiary structure (three-dimensional conformation) and quaternary structure are maintained by van der Waals forces, hydrogen bonds, ionic bonds, and disulfide bonds.

Complete denaturation disrupts bonding in all other levels of protein structure except primary structure.

**Time spent:**0 sec

**Subject:**

Biochemistry

**QID:**402089

**System:**

Amino Acids and Proteins